GROUP WORK PROJECT #01
GROUP NUMBER: 6242

MScFE 600: FINANCIAL DATA

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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Note: You may be required to provide proof of your outreach to non-contributing members upon request.

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Step 1: Collateral Related Risks

	Financing Challenges	Collateral Challenges
Money at a fixed rate for an unsecured purchase	1. Lack of collateral increases downside risk in the event of a buyer default. 2. When interest rates continue to rise, the lender is locked into a fixed rate receivable, which means they may lose out on potential revenues if they sell a loan at a higher or even variable rate. 3. Similarly to interest rates, with fixed pre-determined payments, inflation can eat away at gains for the lender. 4. There is still a credit risk the lender must be aware of that could cause delays in payments or receiving lower payments than expected. Since the loan is unsecured, the credit risk is higher.	
Money at a floating rate for a secured purchase	1. If interest rates rise and the borrower is unable to pay in full, this could affect the lender's receivables. 2. If interest rates fall, the lender would be losing out on potential revenue they could have obtained from using a fixed-rate product. 3. Fluctuations in the value of the collateral can prove risky to the lender. The downside could put a larger part of the loan at risk in case the buyer wishes to walk away without facing recourse. 4. Buyers can always default	

	on the loan due to a myriad of adverse conditions. Assuming the loan is for the secured purchase of a home, adverse conditions like fire, flood, earthquake etc can make the borrower walk away leaving the worthless collateral. 5. Converting the collateral to cash may not be easy or direct.	
Money at a fixed rate for a business for a construction loan	1. The value of the project may decrease over time, putting more money at risk. 2. Converting the loan to cash quickly may be difficult for the lender, especially depending on market conditions and the completion status of the construction project. 3. There is a risk of the project's completion status not happening during the initially proposed duration. 4. There is also risk from the contracting parties, which can go bankrupt, perform poorly, and cause disputes, which can impact the project's success and the ability of the borrower to pay off the loan. 5. Interest rate risk: Since lending is at a fixed rate, increase in overall interest rates would reduce the value of the future fixed payments of principal and interest on the loan. 6. Construction loans are usually not collateralized so the lender faces default risk.	

Publicly traded Equity		1. Accurately valuing collateral can be difficult. 2. Corporate actions can affect the price movements of public equity. This includes issuing dividends, making shareholder announcements, merging with another company, acquiring a company, etc. 3. The lender also faces collateral reinvestment risk. Lenders of securities usually reinvest the collateral (cash) to generate additional yield. If the cash is invested in illiquid or volatile assets, then they face the risk of losing the collateral value.
Publicly traded bond		1. Price transparency and valuation of the underlying bond being lent are important. 2. Bond issuers could default on bond payments leaving the lender in the lurch. 3. Changes in interest rates can impact the bond's value. Longer-term bonds are more sensitive to these changes. 4. If the bond is downgraded or the issuer experiences other specific corporate actions which make it difficult for them to repay the loan, this can change the bond's value.
An illiquid security: private business shares		1. For an illiquid security, one of the more prominent risks is the ability to find a buyer for the asset. Low liquidity can lead to volatile or depressed prices. 2. Valuing an illiquid asset that isn't subject to market forces can be difficult.

		3. Business events can impact the valuation of the shares.

Step 2: Statistical Related Challenges

	Volatility Challenges	**Correlation Challenges**
Money at a fixed rate for an unsecured purchase	1. During market downturns, significant default rates as a result of income volatility 2. Volatility in personal expenses for borrowers can lead to defaults or credit risks. 3. Interest rates can impact the ability of the borrower to pay loans and reduce the amount of money they have to service their cards, etc. Additionally, rising interests will squeeze margins.	1. Economic performance as a whole 2. Public markets performance 3. Many borrowers tend to default at the same time because they face similar hardships and challenges. 4. Interest rates can determine the ability to pay. 5. Correlation of unsecured loan and default risk. The probability of default is usually higher when the loan does not involve any collateral.
Money at a floating rate for a secured purchase	1. If rates rise over time, borrowers will find it more difficult to pay back the loans, leading to greater defaults. 2. Economic downturns can decrease the value of collateral such as homes or businesses or, automobiles, etc.	1. Economic performance affects the value of the collateral. 2. Central bank policies regarding interest rates will correlate with lender's pricing and borrowers' defaults. 3. The housing market and automobile market are correlated with the economy at large. 4. Correlation of interest rate and default risk.

		Under variable-rate mortgage, when interest rates increase, the monthly payments also increase, making it more difficult for the borrower to meet the payments.
Money at a fixed rate for a business for a construction loan	1. Changes in economic conditions can lead to interest rate volatility. At a fixed rate, if interest rates increase, the lender will lose out on revenue. 2. Fluctuations in the price of construction materials and labor can add an element of unpredictability.	1. Property values (the collateral) are generally correlated with broader market trends. 2. There may be a correlation of borrower creditworthiness based on region, sector, or other breakout.
Publicly traded Equity	1. Stock price volatility can increase during periods of securities lending.	1. Correlation with market volatility tends to be positive. More traders will try to borrow more shares, potentially amplifying market movements.
Publicly traded bond	1. Bond prices are highly sensitive to changes in interest rates. In a market where interest rates experience changes or provide the risk/possibility of big movements, bond prices can fluctuate significantly. 2. Expectations of inflation can impact bond prices	1. Correlation with interest rates 2. Correlation between issuer's credit risk and the price of the bond 3. There is an uncertain correlation between equity markets which depends on prevailing economic conditions which may affect bond and equity market investors in similar or opposite ways.
An illiquid security: private business shares	1. There can be uncertainty and	1. Correlated with broader market

	disagreement around valuations, causing volatility in price. 2. Limited liquidity can also lead to volatility in pricing	conditions and industry performance depending on the sector the business operates in and its own performance.

Step 3: Identifying Data

	Questions
Money at a fixed rate for an unsecured purchase	1) Data Type: Interest Rates 2) Data Processing: Levels 3) Data Frequency: Monthly (credit reporting cycle in the US is monthly) 4) Data Class: Credit 5) Data Source: Credit Bureaus 6) Data Variety: Actual Data
Money at a floating rate for a secured purchase	1) Data Type: Floating Interest Rates 2) Data Processing: Raw prices 3) Data Frequency: Daily 4) Data Class: Real Estate 5) Data Source: Mortgage Lenders 6) Data Variety: Actual Data
Money at a fixed rate for a business for a construction loan	1) Data Type: Land Values 2) Data Processing: Raw Prices 3) Data Frequency: Daily 4) Data Class: Real Estate 5) Data Source: Government agencies or Realtor websites 6) Data Variety: Estimated data
Publicly traded Equity	1) Data Type: Stock prices 2) Data Processing: Raw prices 3) Data Frequency: Daily closing prices 4) Data Class: Equity 5) Data Source: Brokers or Financial news 6) Data Variety: Trade data
Publicly traded bond	1) Data Type: Bond yields

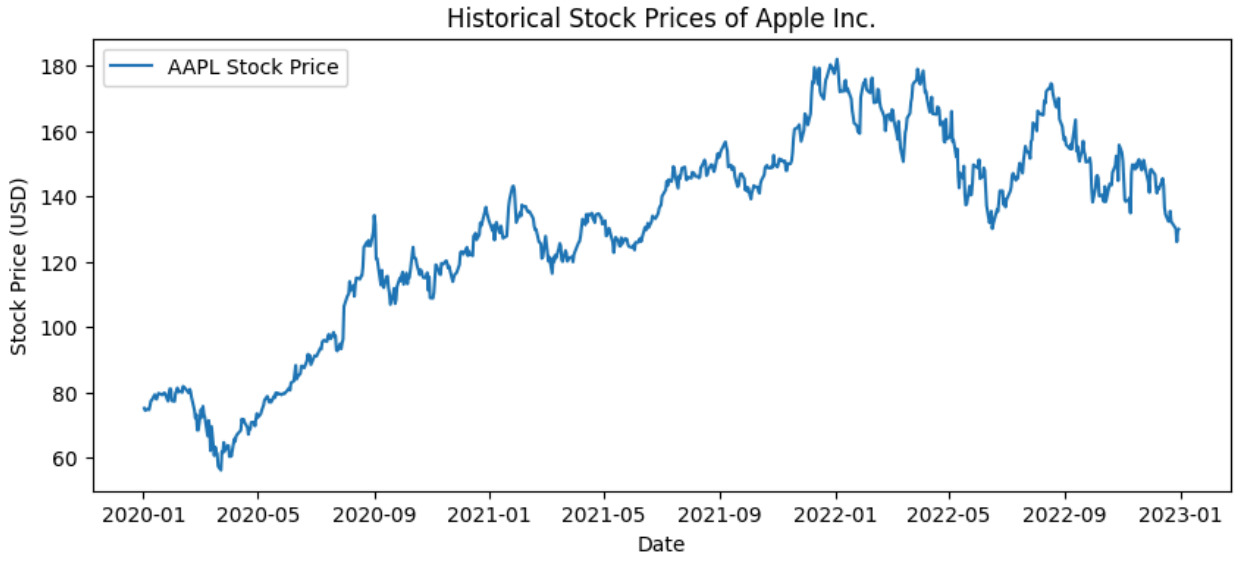
	2) Data Processing: Raw Yields 3) Data Frequency: Daily 4) Data Class: Fixed Income 5) Data Source: Bond markets, credit rating agencies 6) Data Variety: Trade data
An illiquid security: private business shares	1) Data Type: Transaction prices 2) Data Processing: Raw prices 3) Data Frequency: Infrequent (as events occur) 4) Data Class: illiquid securities 5) Data Source: Private Equity firms 6) Data Variety: Actual data

Step 4: Obtaining the Data

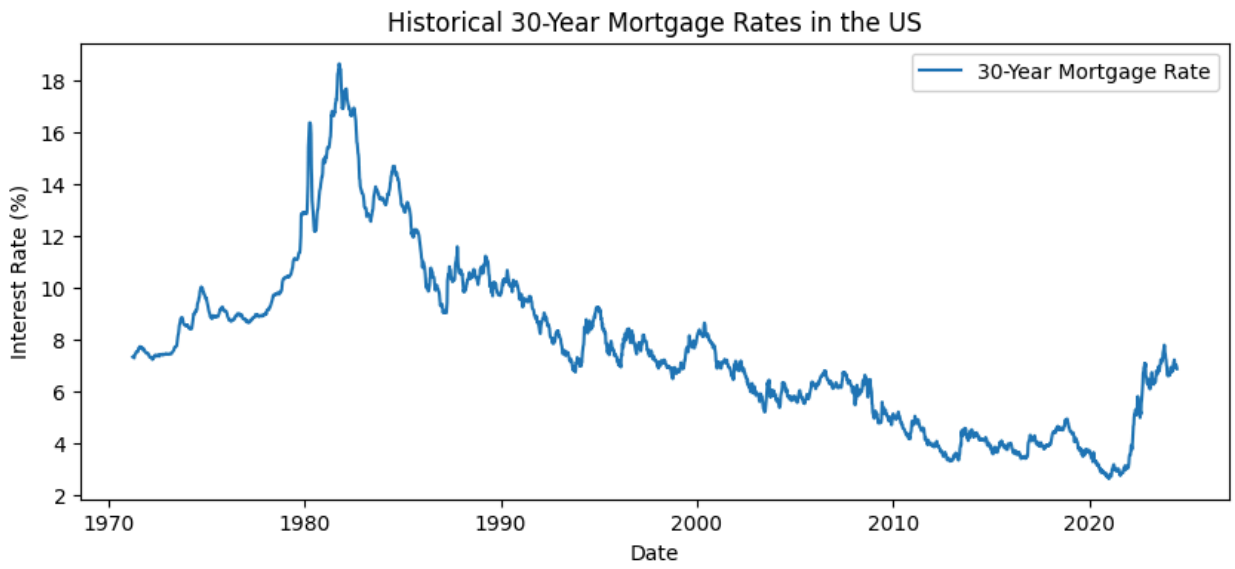
In order to maintain simplicity, we will utilise sample data from Yahoo Finance and other publicly available sources.

Step 5. Perform exploratory data analysis

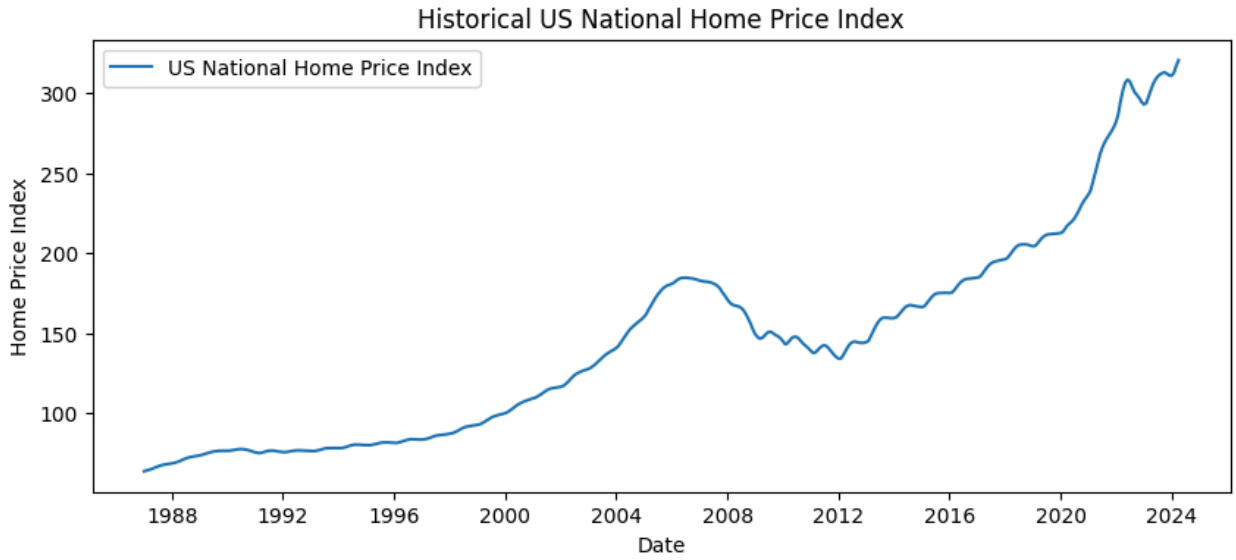
After executing the code, it produces visual representations of historical stock prices, interest rates, home prices, and the distribution of credit ratings. These visualisations provide valuable insights into various volatility and correlation challenges of various lending scenarios.



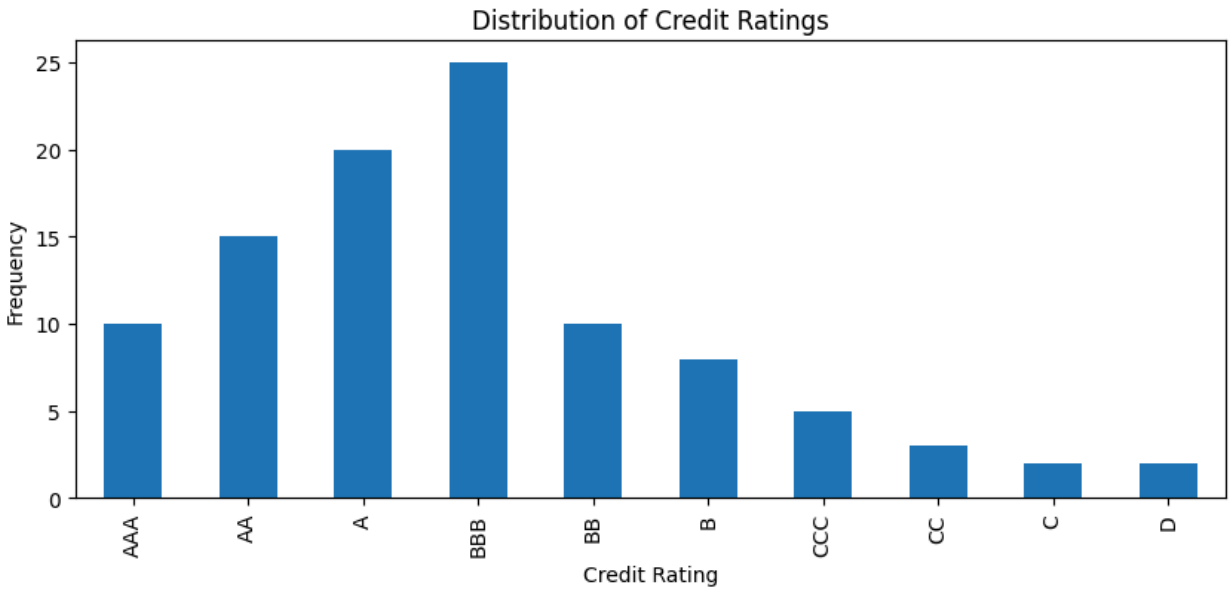
Insight: Apple Inc.'s stock price volatility has displayed significant peaks and troughs, suggesting potential risks for equity investments.



Insight: Fluctuations in US 30-year mortgage rates suggest potential volatility in borrowing costs for floating-rate mortgages.



Insight: The US National Home Price Index has exhibited a general uptrend, indicating a rise in collateral value for secured loans, albeit with intermittent volatility.



Insight: Credit ratings are concentrated in the mid to high range (BBB to AA), signifying diverse degrees of credit risk among borrowers.

Step 6: Describe How the Data Can Help to Meet the Challenge

Interest Rates Data:

- **Challenge:** Volatility in floating-rate mortgages.
- **Insight from Data:** Analysis of historical interest rate patterns enables predicting future rate movements and assessing borrower default risk. Visualising historical interest rates allows us to discern rate fluctuations and detect patterns that may influence future rates.

Stock Prices Data:

- **Challenge:** Volatility in publicly traded stocks.
- **Insight from Data:** Analysis of stock price trends and volatility measures offers valuable insights for evaluating the risk associated with securities lending. By reviewing historical stock prices, we can identify periods of heightened volatility, indicating increased lending risk against these securities.

Reference

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